Practice Exam - 1

Question 117 of 117

A 25-year-old female former college athlete presents for evaluation of her second episode of chest pain and palpitations occurring within two months of each other. The pain occurs both at rest and with exertion, and there is some relief with leaning forward. She has a history of childhood asthma, and about three months ago she had an upper respiratory tract infection. She is not taking any medications. The patient has never had syncope or limitations to exercise. A 54-year-old uncle experienced sudden cardiac arrest while mowing the lawn several years ago. The patient's evaluation is notable for an electrocardiogram demonstrating normal sinus rhythm, normal voltage, and T wave inversions in leads V1 and V2. Serum cardiac troponin I is 0.98ng/dl. Transthoracic echocardiography shows low normal biventricular systolic function without pericardial effusion.

Cardiac magnetic resonance imaging demonstrates biventricular late gadolinium enhancement with a right ventricular ejection fraction of 45% and left ventricular ejection fraction of 48%.

What is the highest yield next diagnostic test for this patient?

A. Cardiac fluorodeoxyglucose-position emission tomography

✗ **B.** Right ventricular endomyocardial biopsy

C. Coronary angiography

✓ **D.** Genetic testing

Commentary References

Testing Point: The Board-certified AHFTC specialist should recognize the indications for genetic counseling and testing.

Answer: D. Genetic counseling and testing. Arrhythmogenic inherited cardiomyopathies (ACM) should be recognized by the AHFTC as a cause for otherwise unexplained cardiomyopathy. This patient has a myopericarditis like syndrome, with chest pain, elevated troponin and abnormal findings on cardiac magnetic resonance imaging. With the patient's mild biventricular cardiomyopathy, one key component of her history that should not be ignored is the family history of sudden cardiac death in an uncle. A 3-generational family history should be performed in any patient presenting with cardiomyopathy. Additionally, she has T wave inversions in V1 and V2, RV systolic dysfunction and a history of competitive athletics the latter of which is a known risk factor for ACM progression. Arrhythmogenic inherited cardiomyopathies are increasingly recognized as having an inflammatory or myocarditis like presentation particularly when due to gene mutations in the desmosomal genes such as DSP. Appropriate referral for genetic counseling and testing can help identify inherited cardiomyopathies, and thus inform arrhythmic as well as family member risk.

Answer choices A-C may all be considered as part of the diagnostic work up in any patient with new onset cardiomyopathy and suggestive features however there are a few caveats to consider. Cardiac fluorodeoxyglucose-positron emission tomography imaging can identify cardiac inflammation, for example in the diagnosis of cardiac sarcoidosis. In this patient, it is less likely to yield critical information than some other choices. Endomyocardial biopsy would be indicated as the next step if the patient presented with hemodynamic compromise and/or documented ventricular arrhythmia or conduction disease specifically to diagnose more fulminant forms of myocarditis such as giant cell myocarditis. The history of an upper respiratory tract infection 3 months prior is a distractor for the risk of myocarditis but it is critical not to ignore the information provided by the family history. Lastly, coronary angiography is likely to be of low yield in this young patient with atypical chest pain features for acute coronary syndrome. It is reasonable however to rule out coronary artery disease, but this could be considered via coronary computed tomography instead given her low risk profile, pre-test probability and adequately controlled heart rate.